

LoraWiFi Drone Detector

User manual and description

Introduction

Welcome to the LoraWiFi drone detector. This device listens with a powerful microphone, analyses the sound it hears, and sends an alert if it detects a drone's sound. The alert is then sent on a LORA radio communication line. The counterpart of the drone detector device listens to these messages.

Deployment

The drone detectors can be deployed as a line or array of detectors. For instance protecting a physical border. But if an area or an installation needs to be protected, any configuration which covers the target area is acceptable. The microphone's angular characteristic is approximately a 45 degree cone.

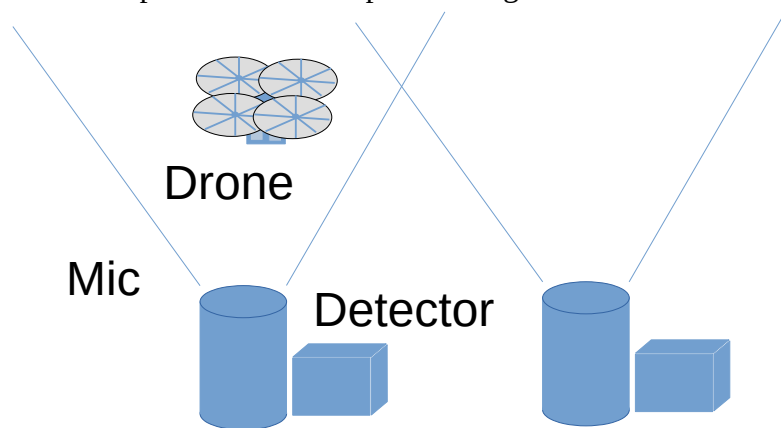


Table of Contents

Introduction:.....	1
Deployment:.....	1
Table of Contents.....	1
Innovation:.....	2
Functional description:.....	2
Power supply:.....	3
Alert Communication:.....	3
Communication robustness:.....	3
Configuring the LoraWiFi Drone Detector unit:.....	5
The WiFi Communication section.....	5
The LORA configuration section.....	7
Explanation of LORE configurable parameters:.....	7
Main Screen:.....	9
The Monitoring section.....	9
The Detection configuration section.....	10

FCC numbers.....	10
Summary:.....	10

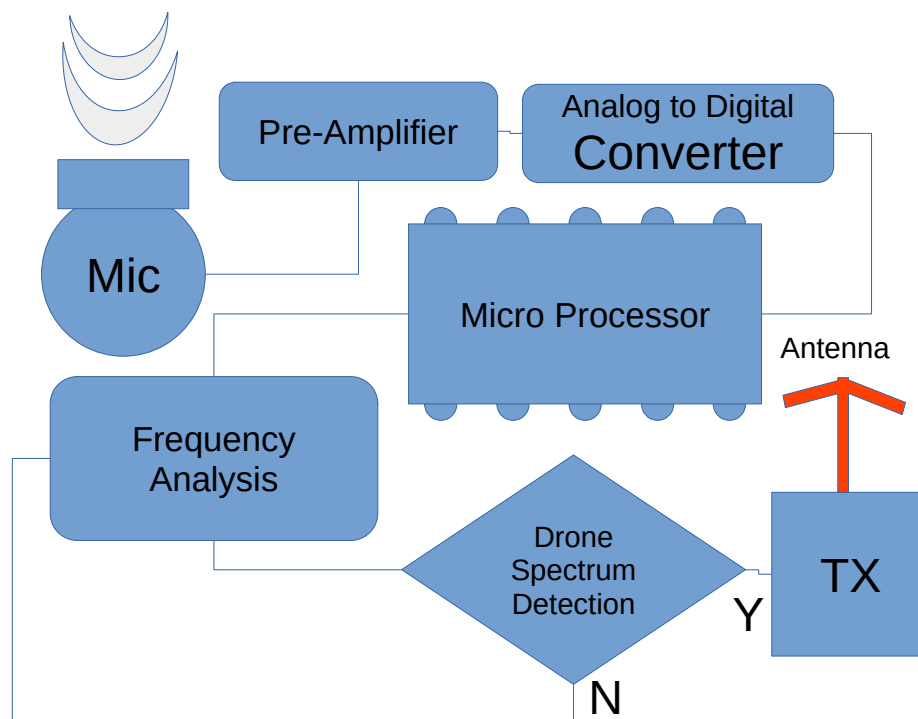
Innovation

The LoraWiFi drone detector contains a state of the art signal analysis algorithm. The algorithm analyzes the incoming sound spectrum, and identifies the dominant frequency components. These frequency components are then compared to a reference set of frequency components. When a match is detected, the drone detector sends the alert signal.

The challenge was to run the frequency analysis on a low power micro-controller. As it has significantly lower processing power than a real microprocessor, extreme innovation was necessary. Happy to announce, that we successfully invented a new type of signal analysis that fulfills the task perfectly. The tests thus far prove excellent performance. The drone noise is successfully and accurately distinguished from many other signals. For instance motorcycle engine noise is not detected as a drone noise. The detection is so accurate, that we venture that the category of drone can also be identified with this algorithm. The false positive ratio and the false negative ratio of a single unit is better than 15 to 1. Multiple units improve this ratio cumulatively.

Functional description

The drone detector is built on top of a state of the art consumer micro-controller, an ultra sensitive microphone and a low power, long range (LORA) transmitter.



Please note that the chart above is an oversimplified version of the operational logic.

Power supply

The device can be powered from a 5V standard USB connector. This can be then extended to any power arrangement; for instance a solar panel powered battery solution. The last 25 messages are stored in non volatile memory, they are preserved across power cycles.

Alert Communication

The alert is communicated with a LORA radio. This operates in the 433 MHz range, and is license free. The specified range of the LORA radio is from 500 meters (in built up areas) to 10 kilometers (line of sight). The practical range is about 300 meters to 5 kilometers.

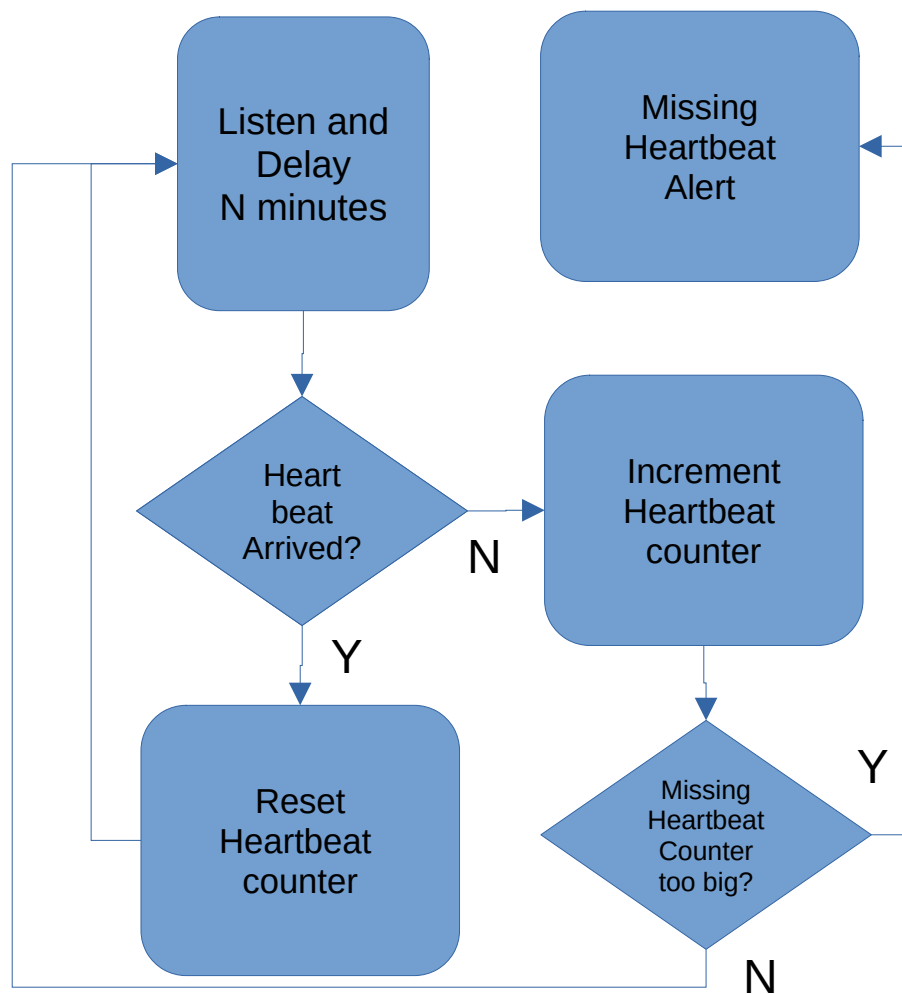
The alert itself can be freely specified, any text up to 230 characters. We recommend as short as possible. For example, the alert text can be: “Alert: Station 34”

The station identification is an essential part of the alert, as it identifies the source of the message.

Communication robustness

The LoraWiFi Drone detector can send a heartbeat, signaling the receiving station that everything is OK. This confirms that the radio link is operational, that the detector is listening, and everything is in working order. If and when the heartbeat is missing, there is something not operational or jammed. In that case, appropriate measures can be taken by the operator of the drone detector system.

The heart beat string can be specified, any text up to 230 characters. We recommend as short as possible. For example, the heart beat text can be: “OK, Station 34” The station identification may be an essential part of the heartbeat, as it identifies the source of the heartbeat message.



Please note that the flowchart above is an oversimplified version of the required logic.

Any LoraWiFi Drone detector can be used as a monitoring station as well. The occurrence of the alert can be communicated in several ways. One of the most diverse of those methods is a terminal string. The LoraWiFi USB can be connected to a computer (or cell phone), and a terminal emulation program can read the output of the drone detector. The alert is delivered as a text, which is interpretable. These text strings are also configurable, which allows easy integration with other systems.

The second monitoring facility is a WiFi transmission. This essentially acts as a LORA to WiFi router. Whatever comes in via the LORA channel, is replicated to WiFi. The WiFi signal than can be interpreted as desired.

The third monitoring facility is a simple colored LED. The LED is lit up in bright red color upon detection.

The fourth monitoring facility is a small buzzer. This sounds briefly on detection.

The missing heartbeat signal can be communicated by the same method. The difference is the actual value of the terminal string, or the WiFi repeat signal, or the color of the LED, which is lit up orange.

Configuring the LoraWiFi Drone Detector unit

In order to configure the device, one needs to connect to the WiFi device interface presented. The name of the device is 'LoraWiFi-NNNN' where the 'NNNN' is the last 4 digits of the device ID. (mac address) For example, one of the development board's name was: 'LoraWiFi-0567' (without the quotes). The default password to connect to the board is '12345678' (without the quotes). After the connection is established, one may connect to the device's browser. In the browser's URL bar enter:

http://192.168.4.1

The device's home page should be presented, which is pictured below:

LoraWiFi Drone Detector

Current Drone Detection status: Active

No alerts yet ...

Past Alerts:

Trench: (0-65534) ☐ Show all trenches

Send:

Count: 0
Max 230 Characters
Ctrl-Enter to submit

Status: Last RSSI: Last devi: Last PPM:

[Main Settings](#) [WiFi Settings](#) [LORA Parameters](#)
[Alert Settings](#) [Monitoring Settings](#) [Detection Parameters](#)
[Quick Start Manual](#) [Go to Home Page](#) [Reboot Device](#)

comstatcomstatcomstatcomstatcomstat

ID (Main) (Mac Address): macmacmacmacmacmacmac
Current Boot Count: bcountbcount
Software Version / Build Date: softversoftversoftversoftversoftversoftver
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The device configuration consists of three main sections.

1. The WiFi Communication section
2. The Detection section
3. The Monitoring section

The WiFi Communication section

This section allows one to configure the WiFi wireless radio subsystem related items. The first screen configures the WiFi host name and password. This is standard configuration item, the screen below is mostly self explanatory.

LORA WiFi Transceiver

The following fields determine the Station Name and Password of this LORA WiFi. The device will appear on your WIFI network by this name. For safety and security, the LORA WiFi must be configured with a password. The password needs to be at least **8 characters** or longer. Please avoid using space, punctuation marks and non ASCII characters.
You may use the _ (underscore) - (minus) sign + (plus) sign.

New Web Login Credentials:

Station name:

Station password:

Repeat password:

Save Web Configuration (Alt+Shift+c)

| WiFi FCC-ID: 2AC7Z-ESPS2MINI1 | LoRA FCC-ID: 2BCKU-RA0212 |

[Main Settings](#)
[Quick Start Manual](#)

[WiFi Settings](#)
[Go to Home Page](#)

[LORA Parameters](#)
[Reboot Device](#)

ID (Main) (Mac Address): macmacmacmacmacmacmac
Current Boot Count: bcountbcount
Software Version / Build Date: softversoftversoftversoftversoftver
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The LORA configuration section

The second screen configures the LORA subsection. The screen below depicts the configuration items. The defaults of the communication parameters can be restored by pressing on a 'Reset LORA parameters' button.

LORA WiFi Transceiver

The following fields determine the LORA transmit / receive configuration. These configurations must match between the sender and receiver units. The parameters are range contained. The frequency parameter is a floating point number multiplied by megahertz. The bandwidth parameter is a floating point number multiplied by kilohertz. Please note: for communication stability the lowest recommended bandwidth is 40+ kHz. (This is due to the LORA Board's crystal inaccuracy and / or instability) See the 'Tune RX offset' section for more information.

LORA parameters:

LORA Spread Factor: (6 - 12)

spreadspread

LORA Bandwidth: (10 - 500) kHz

bandbandbandband

Optional (advanced) LORA parameters:

LORA TX power (1-15):

txpowtxpowtxpow

LORA Frequency: (433.05 - 434.79) MHz

freqfreqfreqfreqfreqfreq

[\(Channel Table\)](#)

Save LORA Configuration

Tune LORA RX (receive) offset:

This field tunes the LORA receive frequency offset. This step is optional, and it may be necessary if communication is difficult at lower bandwidths. The tuning value can be derived from the Main LORA page's 'devi' field. The tuning value should be set to decrease the 'devi' field on the main page as close to zero as possible. The tuning value approximates the -devi/2. (minus (half devi)). For instance if devi is -10000 the tuning value should be 5000.

LORA RX tuning (+-) (0-10000):

rxtunerxrtunerxtune

Tune RX Frequency offset.

Reset LORA parameters:

The button below resets the LORA parameters to manufacturer's defaults. This may be necessary if communication configuration becomes out of sync with the target. Please remember to change these parameters for increased distance and increased privacy. Note that both the sender and receiver needs identical parameters to communicate. This action also resets the RX tuning.

Reset LORA Parameters

| WiFi FCC-ID: 2AC7Z-ESPS2MINI1 | LoRA FCC-ID: 2BCKU-RA0212 |

Explanation of LORE configurable parameters:

The LoraWiFi system utilizes a SEMTEC LORA device. The LORA device's specification suggest an 8-10 kilometer (line of sight) range. Under ideal conditions, this can be measured and proven with this implementation, and it delivers according to specification. Using the device in a non line of sight configuration, the range is significantly less. For instance using in high rise building area, the range can be as little as 500 meters. The chip manufacturer's web site contains additional information on LORA deployment.

Parameter	Range of Values	Notes
Spread factor	6 - 12	Determines how many RF chirps per communication unit. Higher value results in a longer range and longer transmission time. Default: 10
Bandwidth	10 kHz – 500 kHz	The modulation width. Lower values result in longer range, and longer transmission time. Default: 50 kHz
TX power	1 - 15	The TX power. Higher values result in longer range. Values of 12 and above switch on the power boost feature. This feature turned on may cause communication error at very short

Center Frequency 433.05 to 434.79 MHz

distances.

Default: 12

This is the center frequency of the LORA transmission / reception. The value can be set to any frequency that is between the specified limits. It is recommended to follow the LORA channel allocation, however it is not required. This may give additional communication robustness when specifying arbitrary center point.

Default: 433.375 (Channel 1)

Calibration Offset -20000 Hz to +20000 Hz

The reception frequency is calculated as the transmission frequency + calibration offset. This allows compensation for crystal frequency offsets.

Default: 0

The LORA chip maker has a range calculator, at:

<https://www.semtech.com/design-support/lora-calculator>

Submitting the default settings (after set to max power) of the LoraWiFi Drone Detector to the chip maker's site results in:

Summary ▶

Timing Results

Time on Air: 495.61 ms

Symbol Time: 16.38 ms

Effective Data Rate: 488.28 bps

Total symbol: 30.25 Symbols

Preamble Duration: 131.07 ms

RF Performance

Radio Consumption: 5.02 uAh

Link Budget: 155.0 dB

Max Crystal Tolerance: 17.1 ppm

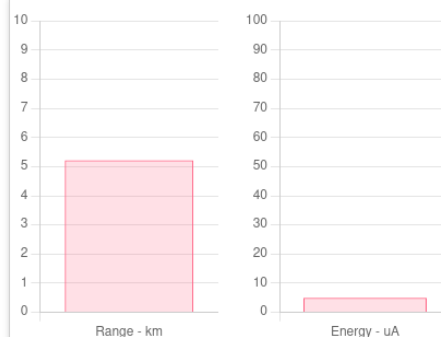
Max Crystal Tolerance: 15625.0 Hz

Receiver Sensitivity: -135.0 dBm

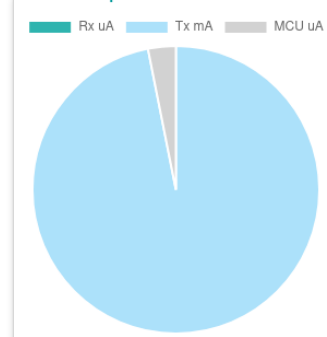
Device Tx Consumption: 125000 uA

Device Rx Consumption: 10.8 mA

Performance



Consumption



Please note, that this assumes a line of sight communication.

Main Drone Detector Screen

The device

LoraWiFi Drone Detector

Main Page

Current Drone Detection status: Active

Past Alerts:

No alerts yet ...

Trench:

trenchtrenchtrench

(0-65534)

☐ Show all trenches

Send:

Count: 0
Max 230 Characters
Ctrl-Enter to submit

Send

Status:

Idle.

Last RSSI: 0

Last devi: 0

Last PPM: 0

[Main Settings](#)

[Alert Settings](#)

[Quick Start Manual](#)

[WiFi Settings](#)

[Monitoring Settings](#)

[Go to Home Page](#)

[LORA Parameters](#)

[Detection Parameters](#)

[Reboot Device](#)

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ID (Main) (Mac Address): macmacmacmacmacmacmac

Current Boot Count: bcountbcount

Software Version / Build Date: softversoftversoftversoftversoftver

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The Detection configuration section

The LoraWiFi drone detection can be configured on this screen.

The Monitoring section

The monitoring section can configure

FCC numbers

This device inherits two FCC authorizations, one from both radio units. The unit's FCC numbers:

WiFi FCC-ID: 2AC7Z-ESPS2MINI1
LoRA FCC-ID: 2BCKU-RA0212

Summary:

Summary